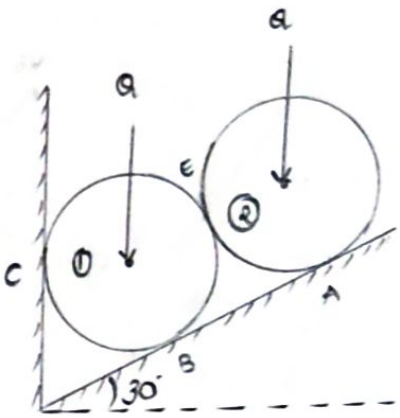
1, Two identical rollers, each of weight 100N, are supported by an inclined plane and a vertical wall as shown in fig. Assuming smooth surfaces, find the reactions at the points of support A, B and C.

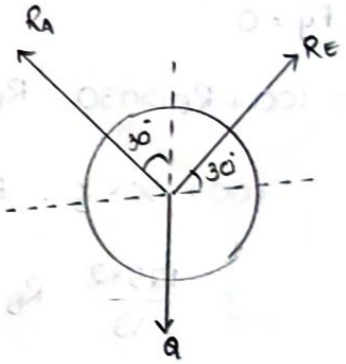


Sol:- Given that,

The weight of two rollers = 100N

From the free body diagram of roller ② we can apply lami's theorem.

Roller ① acts as a support to roller ②



Lami's theorem,

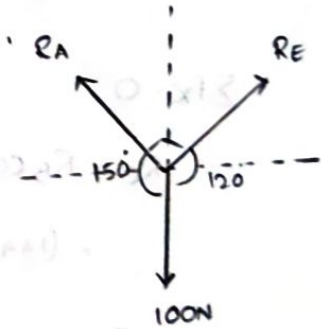
$$\frac{100}{\sin 90^\circ} = \frac{R_A}{\sin 120^\circ} = \frac{R_E}{\sin 150}$$

$$\frac{R_A}{\sin 120^\circ} = 100$$

$$R_A = 100 \sin 120^\circ$$

$$R_A = 100 \times \frac{\sqrt{3}}{2}$$

$$R_A = 86.6025 N$$

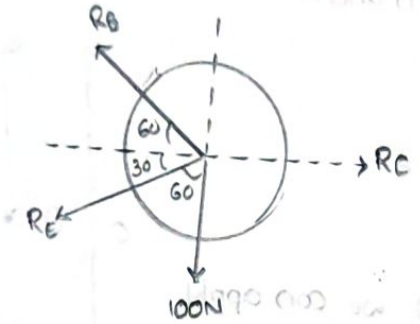


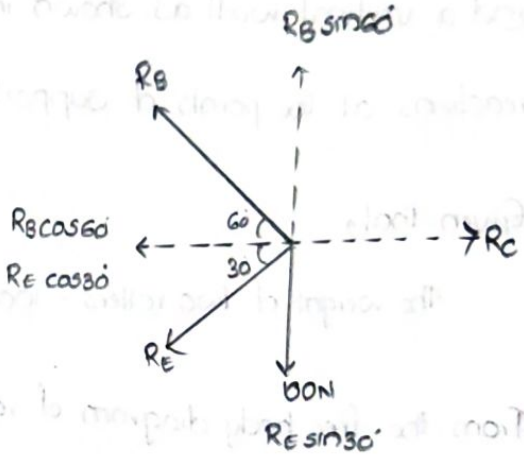
$$\frac{R_E}{\sin 150^\circ} = 100$$

$$R_E = 100 \sin 150^\circ$$

$$R_E = 50N$$

Now consider free body diagram of ①





$\Sigma F_y = 0$

$100 + R_E \sin 30° = R_B \sin 60°$

$100 + 50 \times \frac{1}{2} = R_B \left(\frac{\sqrt{3}}{2}\right)$

$\frac{125 \times 2}{\sqrt{3}} = R_B$

$R_B = 144.34\ N$

$\Sigma F_x = 0$

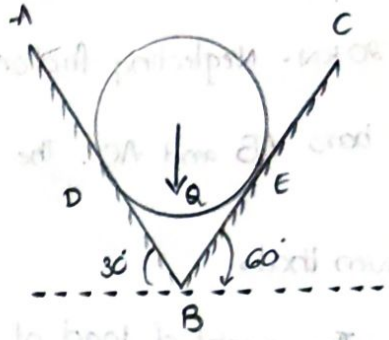
$R_C = R_B \cos 60° + R_E \cos 30°$

$= (144.34)\frac{1}{2} + 50 \times \sqrt{3}/2$

$R_C = 115.47$

$R_C = 115.47\ N$

2, A ball of weight Q = 12N rests in a right angled trough, as shown in fig. Determine the forces exerted on the sides of the trough at D and E if all surfaces are perfectly smooth.

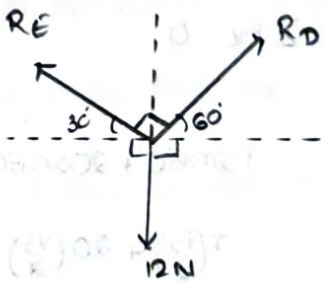


Sol: Given that,

ball of weight Q = 12N

from free body diagram we can apply lami's theorem.

$$\frac{12}{\sin 90^\circ} = \frac{R_D}{\sin 120^\circ} = \frac{R_E}{\sin 150^\circ}$$



$$\frac{12}{1} = \frac{R_D}{\sin 120} = \frac{R_E}{\sin 150^\circ}$$

$$R_D = 12 \sin 120$$

$$R_D = 12\left(\frac{\sqrt{3}}{2}\right)$$

$$R_D = 6\sqrt{3}$$

$$R_D = 10.392N$$

$$R_E = 12 \sin 150^\circ$$

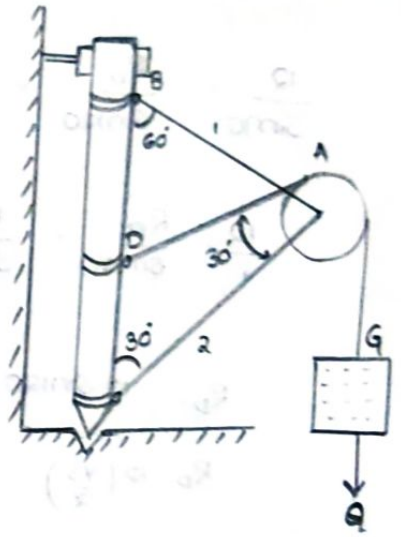
$$= 12\left(\frac{1}{2}\right)$$

$$R_E = 6N$$

3) A pulley A is supported by two bars AB and AC which are hinged at points B and C to a vertical mast EF. Over the pulley hangs a flexible cable DG which is fastened to the mast at D and carries at the other end G a load G = 20 KN. Neglecting friction in the pulley, determine the forces produced in the bars AB and AC. The angles between the various members are shown in fig.

Soln Given that,

The weight of load at G = 20KN



from free body diagram,

$\Sigma F_x = 0$

$T\sin 60° + 20\sin 60° = C_p \sin 30°$

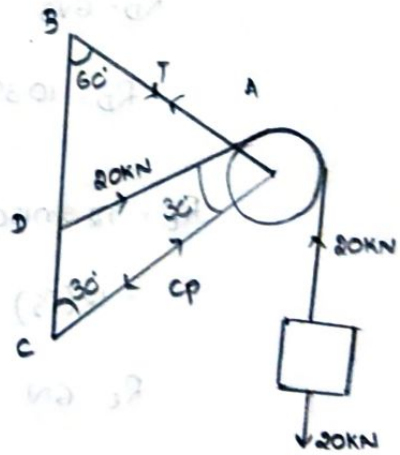
$T\left(\frac{\sqrt{3}}{2}\right) + 20\left(\frac{\sqrt{3}}{2}\right) = C_p\left(\frac{1}{2}\right)$ — (2)

$\Sigma F_y = 0$

$20 + 20\cos 60° = T\cos 60° + C_p\cos 30°$

$20 + 20\left(\frac{1}{2}\right) = T\left(\frac{1}{2}\right) + C_p\left(\frac{\sqrt{3}}{2}\right)$

$30 = T\left(\frac{1}{2}\right) + C_p\left(\frac{\sqrt{3}}{2}\right)$ — (2)



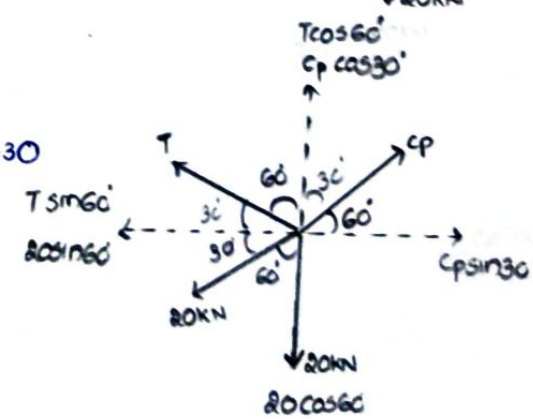
$(1) \times \sqrt{3} = T\left(\frac{3}{2}\right) + 20\left(\frac{3}{2}\right) = C_p\left(\frac{\sqrt{3}}{2}\right)$

$(2) = T\left(\frac{1}{2}\right) + C_p\left(\frac{\sqrt{3}}{2}\right) = 30$

$T\left(\frac{3}{2} + \frac{1}{2}\right) + 20\left(\frac{3}{2}\right) + \cancel{C_p\left(\frac{\sqrt{3}}{2}\right)} = \cancel{C_p\left(\frac{\sqrt{3}}{2}\right)} + 30$

$2T + 30 = 30$

$T = 0$



Substitute $T=0$ in eq ①

$$T\left(\frac{\sqrt{3}}{2}\right) + 20\left(\frac{\sqrt{3}}{2}\right) = C_p\left(\frac{1}{2}\right)$$

$$20\left(\frac{\sqrt{3}}{2}\right) = C_p\left(\frac{1}{2}\right)$$

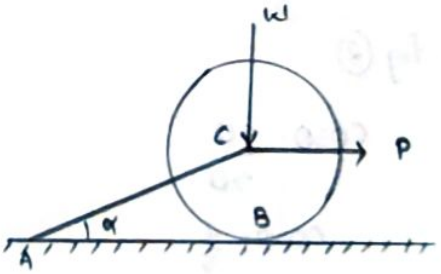
$$C_p = 20\sqrt{3}$$

$$C_p = 34.640N$$

$\therefore$ The force produced in the bar AB $T=0$

The compression produced in the bar AC $(C_p) = 34.640N$

4, A right circular roller of weight 'w' rests on a smooth horizontal plane and is held in position by an inclined bar AC as shown in fig. Find the tension 'S' in the bar AC and the vertical reaction R_b at B if there is also a horizontal force 'P' acting at C.



Sol:- Given that,

the weight of circular roller is 'w'.

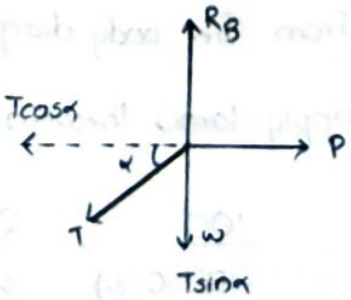
From free body diagram,

$\Sigma F_x = 0$

$P = T\cos\alpha$ — ①

$\Sigma F_y = 0$

$w + T\sin\alpha = R_B$ — ②



$\Rightarrow$ ① $T = P/\cos\alpha$

substitute the value of 'T' in ②

$$w + P\left(\frac{\sin\alpha}{\cos\alpha}\right) = R_B$$

$P\tan\alpha = R_B - W$

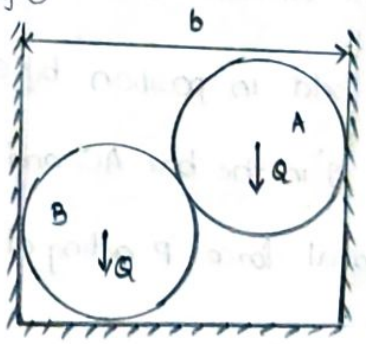
$R_B = W + P\tan\alpha$

$\therefore$ The tension in bar AC $= P/\cos\alpha$

$\therefore$ The reaction at B $(R_B) = W + P\tan\alpha$

5. Two smooth spheres, each of radius r and weight Q, rest in a horizontal channel having vertical walls, the distance between which is b. Find the reactions exerted on the walls and floor at the points of contact A, B and D. The following numerical data are given $r = 25cm$, $b = 90cm$, $Q = 100N$.

Fig: ①



Sol: Given that,

Weight of smooth spheres = 100N

radius of each sphere = 25cm

From fig ②

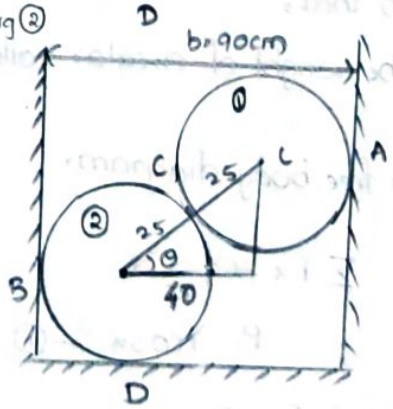
$\cos\theta = \frac{40}{50}$

$\cos\theta = \frac{4}{5}$

$\theta = \cos^{-1}(4/5)$

$\sin\theta = 3/5$

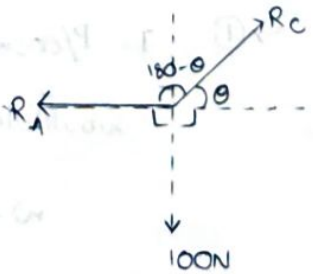
Fig ②



From free body diagram, we can apply lami's theorem.

FBD: ①

$$\frac{100}{\sin(180-\theta)} = \frac{R_A}{\sin(90+\theta)} = \frac{R_C}{\sin 90}$$



$$\frac{100}{\sin\theta} = \frac{R_A}{\cos\theta} = R_C$$

$$100 = R_C \sin\theta$$

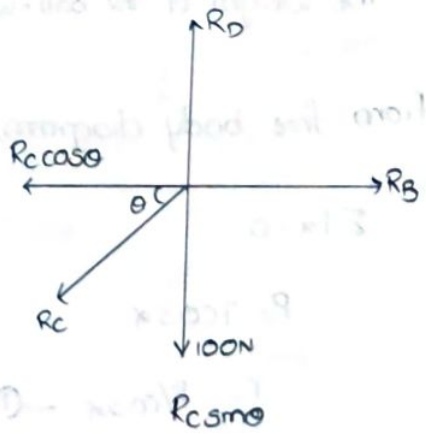
$$R_C = 100 \times 5/3$$

$$R_C = 500/3 \text{ N}$$

$$R_A = R_C \cos\theta$$

$$R_A = \frac{400}{3} \text{ N}$$

FBD of ②



From FBD of ②,

$\Sigma F_y = 0$

$$R_C \sin\theta + 100 = R_D \quad \text{—①}$$

$\Sigma F_x = 0$

$$R_B = R_C \cos\theta \quad \text{—②}$$

② ⇒ $R_B = \frac{500}{3} \times \frac{4}{5}$

$$R_B = \frac{400}{3} \text{ N}$$

① ⇒ $R_D = \frac{500}{3} \times \frac{3}{5} + 100$

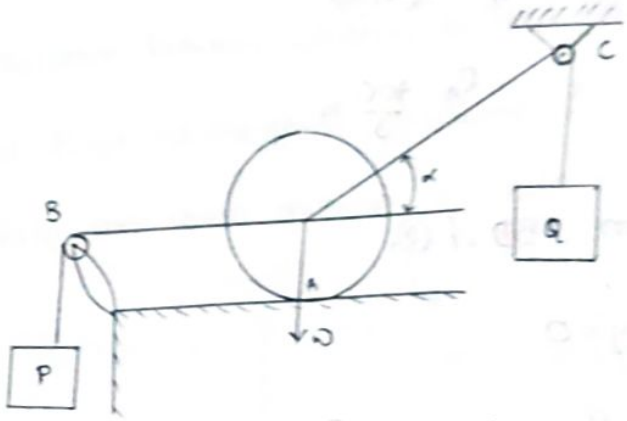
$$R_D = 200\text{N}$$

∴ Reaction exerted at point A = $400/3$ N

Reaction exerted at point B = $400/3$ N

Reaction exerted at point D = 200N

6. A ball of weight W rests upon a smooth horizontal plane and has attached to its centre two strings AB and AC which passes over frictionless pulley at B and C and carry loads P and Q, respectively, as shown in Fig. If the string AB is horizontal, find the angle α that the string AC makes with the horizontal when the ball is in position of equilibrium. Also find reaction R between the ball and the plane.



Sol: Given that,

the weight of the ball = W

from free body diagram,

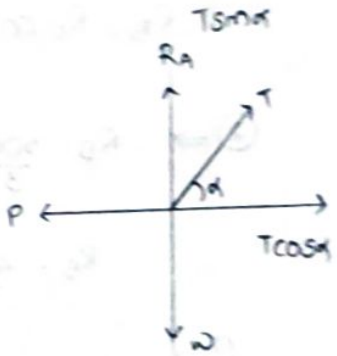
$\Sigma F_x = 0$

$P = T\cos\alpha$

$T = P/\cos\alpha$ —①

$\cos\alpha = P/T$

$\alpha = \cos^{-1}(P/T)$



$\Sigma F_y = 0$

$R_A + T\sin\alpha = W$

$R_A + P\tan\alpha = W$ (from ①)

$R_A = W - P\tan\alpha$

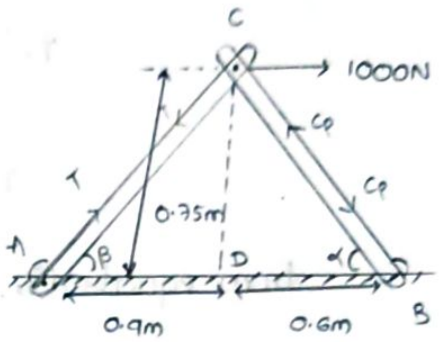
∴ The reaction force $= W - P\tan\alpha$

∴ Angle $\alpha = \cos^{-1}(P/T)$

7, Determine the forces S_1 and S_2 induced in the bars AC and BC in fig. due to the action of the horizontal applied at C. The bars are hinged together at C and to the foundation of A and B.

Soln: In the bar AC tension force is acting

In the bar BC compression is acting.

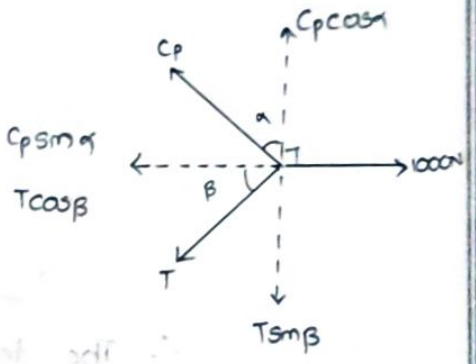


In ΔACD

$$\tan\beta = \frac{0.75}{0.9} = \frac{5}{6} = 0.833$$

In ΔCBD,

$$\tan\alpha = \frac{0.75}{0.6} = \frac{5}{4}$$



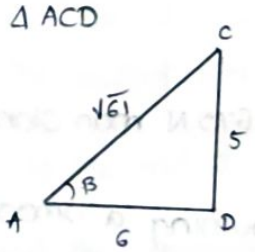
From free body diagram,

$\Sigma F_y = 0$

$Cp\cos\alpha = T\sin\beta$ —①

$$Cp = \frac{T\sin\beta}{\cos\alpha}$$

ΔACD

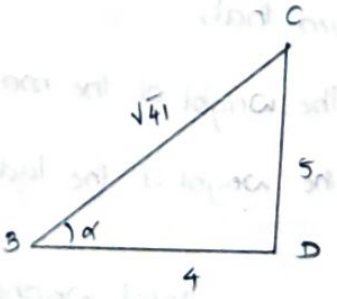


$\Sigma F_x = 0$

$Cp\sin\alpha + T\cos\beta = 1000N$ —②

Substitute Cp value in eq ②

ΔBCD



$$\frac{T\sin\beta}{\cos\alpha}\sin\alpha + T\cos\beta = 1000N$$

$$T\sin\beta\tan\alpha + T\cos\beta = 1000N.$$

$$T\left(\frac{5}{\sqrt{61}}\right)\left(\frac{5}{4}\right) + T\left(\frac{6}{\sqrt{61}}\right) = 1000$$

$$T\left(\frac{25}{\sqrt{61}\times 4} + \frac{6}{\sqrt{61}}\right) = 1000$$

$$T = \frac{4000\times\sqrt{61}}{49}$$

$$T = 637.571\ N$$

from equation ①, $C_p = \frac{T\sin\beta}{\cos\alpha}$

$$= \frac{T\times(5/\sqrt{61})}{(4/\sqrt{41})}$$

$$= 637.571 \times 1.0247$$

$$C_p = 653.319\ N$$

∴ The tension in the bar AC = 637.571N

The compression induced in the bar BC = 653.319N.

8, A 675 N man stands on the middle rung of a 225 N ladder, as in Fig. Assuming a smooth wall at B and a stop at A to prevent slipping, find the reactions at A and B.

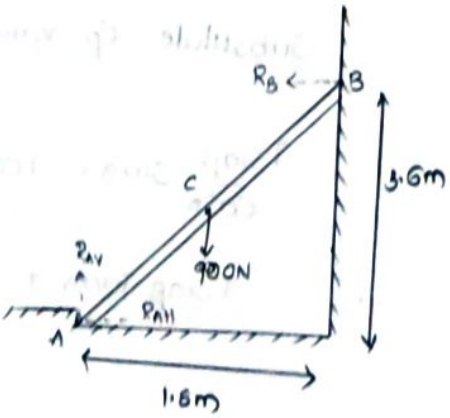
Sol: Given that,

The weight of the man $W_m = 675N$

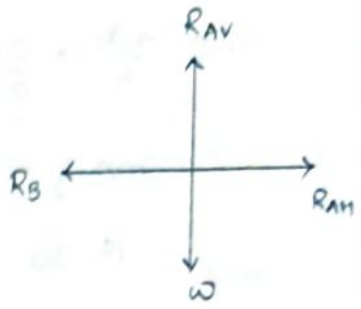
The weight of the ladder $W_L = 225\ N$

Total weight $W_m + W_L = W$

$W = 900\ N$



From free body diagram,



$\Sigma F_x = 0$

$R_B = R_{AH}$

$\Sigma F_y = 0$

$R_{AV} = W$

$R_{AV} = 900N$

The moment at A, $\Sigma M_A = 0$

$R_B \times 3.6 - W \times 0.9 = 0$

$R_B (3.6) - 900 \times 0.9 = 0$

$R_B = 225N$

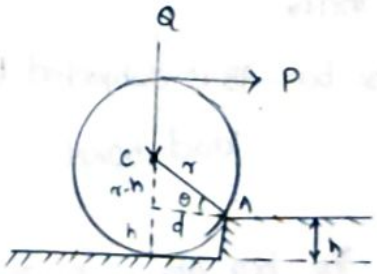
$R_{AH} = 225N$

$\therefore$ Reaction at B = 225

Reaction at A vertically = 900N

Reaction at A horizontally = 225N

9, A roller of radius $r = 0.3m$ and weight $Q = 2000N$ is to be pulled over a curb of height $h = 0.15m$ by a horizontal force P applied to the end of a string wound around the circumference of the roller. Find the magnitude of 'P' required to start the roller over the curb.



Solⁿ Given that,

the radius of cylinder roller (r) = 0.3m

height of the curb = 0.15m (h)

Weight of the roller (W) = 2000N

$\Rightarrow r - h = 0.3 - 0.15 = 0.15$

from fig,

$\sin\theta = \frac{r-h}{h} = \frac{0.15}{0.3} = \frac{1}{2}$

$\sin\theta = \frac{1}{2}$

$\theta = 30^\circ$

$\cos 30^\circ = d/r = \frac{d}{0.3}$

$d = 0.3\sqrt{3}/2$

$d = 0.25980$

Moment at A, $\Sigma M_A = 0$

$w \times d - P(r + r - h) = 0$

$2000 \times 0.25980 = P(2(0.3) - 0.15)$

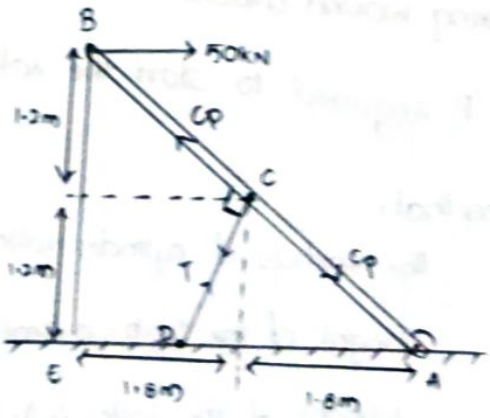
$P = 1154.6666$

$\therefore$ The magnitude of P required = $1154.\overline{66}$

10) A bar AB hinged to the foundation at A and supported by a strut CD is subjected to a horizontal 50 kN load at B, as shown in fig. Determine the nature and magnitude of the force in the strut and also reaction at A.

Soln Given that,

The bar AB is subjected to a horizontal load = 50kN



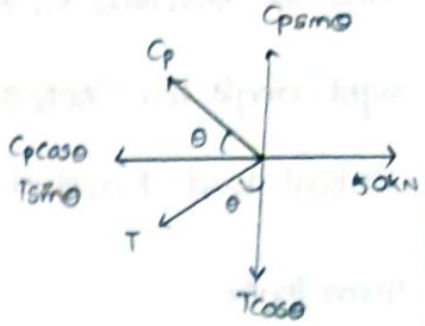
The bar AB is in compression

The strut CD is in Tension.

In ΔACB,

$\tan\theta = \frac{2.4}{3.6}$

$\tan\theta = 2/3 \quad \sin\theta = \frac{2}{\sqrt{13}} \quad \cos\theta = \frac{3}{\sqrt{13}}$



from free body diagram,

$\Sigma F_x = 0$

$C_p\cos\theta + T\sin\theta = 50KN$

$C_p\left(\frac{3}{\sqrt{13}}\right) + T\left(2/\sqrt{13}\right) = 50KN$ —①

$\Sigma F_y = 0$

$T\cos\theta = C_p\sin\theta$

$T\left(3/\sqrt{13}\right) = C_p\left(2/\sqrt{13}\right)$

$T = C_p(2/3)$ —②

Substitute the value 'T' in eq ①

$C_p\left(3/\sqrt{13} + \frac{2}{3}\left(\frac{2}{\sqrt{13}}\right)\right) = 50KN$

$C_p\left(\frac{13}{3\sqrt{13}}\right) = 50KN$

$C_p = \frac{150}{\sqrt{13}}$ KN

$C_p = 41.6025147169$ KN

$T = C_p(2/3)$

$= 41.6025147169(2/3)$

$T = 27.73500981$ KN

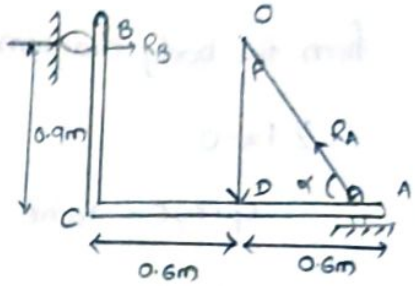
$\therefore$ The tension in the struct CD = 27.73500981 KN

The compression in the bar AB = 41.6025147169 KN

11,

Find the reactions R_a and R_b induced at the supports A and B of the right-angle bar ACB supported as shown in fig., and subjected to a vertical load P applied at midpoint of AC.

Sol: Given that,



Force 'P' is acting vertically downward.

These three forces must be form of a triangle which is similar to triangle AOD.

Therefore,

$$\frac{P}{OD} = \frac{R_B}{0.6} = \frac{R_A}{OA}$$

$$OA = \sqrt{(0.6)^2 + (0.9)^2}$$

$$= 1.081$$

$$\frac{P}{0.9} = \frac{R_B}{0.6} = \frac{R_A}{1.081}$$

$$R_B = \frac{0.6}{0.9} P = \frac{2}{3} P = 0.66P$$

$$R_A = \frac{1.081}{0.9} P = 1.20185P$$

Angle 'α' that makes R_A with the horizontal

$$\tan\alpha = \frac{0.9}{0.6} = 3/2$$

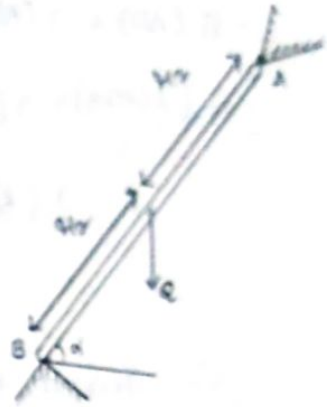
$$\alpha = \tan^{-1}(3/2)$$

$$\alpha = 56.309^\circ$$

∴ The reaction at A 'R_A' = 1.20185P

The reaction at B 'R_B' = 0.66P

12) A prismatic bar AB is hinged at A and supported at B as shown in fig. Neglecting friction, determine the reaction R_b produced at B owing to the weight Q of the bar.



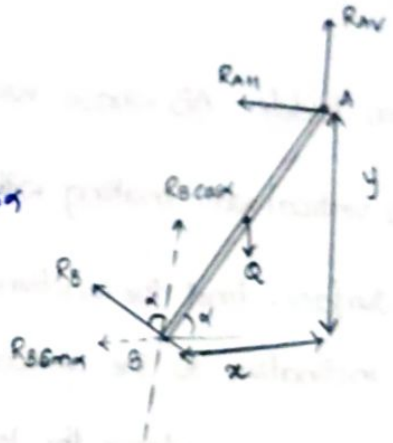
Sol. From free body diagram,

$\Sigma F_x = 0$

$R_B \sin\alpha + R_{AH} = 0$

$\Sigma F_y = 0$

$R_B \cos\alpha + R_{AV} = Q$



Moment at A $\Sigma M_A = 0$

$R_B \cos\alpha \times l\cos\alpha + R_B \sin\alpha\, l \sin\alpha = Q(l/2)\cos\alpha$

$R_B l(\cos^2\alpha + \sin^2\alpha) = Q(l/2)\cos\alpha$

$R_B = \frac{Q\cos\alpha}{2}$

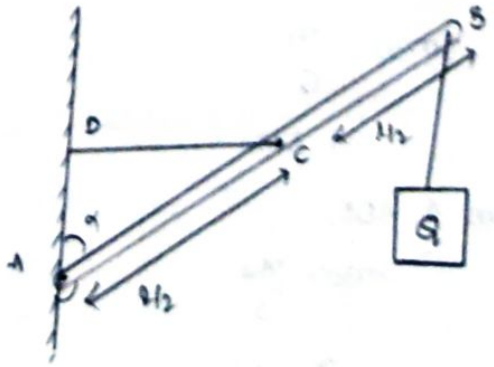
∴ The reaction R_B produced at B = $\frac{Q\cos\alpha}{2}$

13) A rigid bar AB is supported in a vertical plane and carries a load Q at its free end as shown in fig. Neglecting the weight of the bar itself, compute the magnitude of the tensile force 'S' induced in horizontal string CD.

Sol: Given that,

The length of rigid bar = l

The weight hinged at B = Q



From FBD,

Consider $\Sigma M_A = 0$

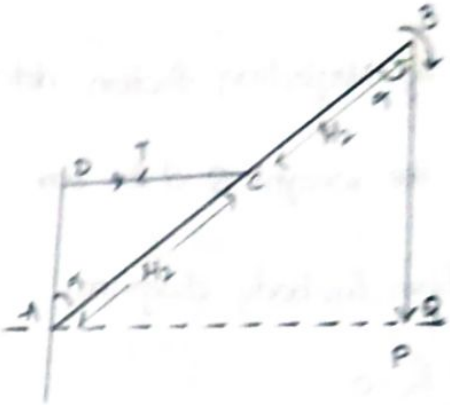
$-Q(AP) + T(AD) = 0$

$-Q(l\sin\alpha) + T(l/2)\cos\alpha = 0$

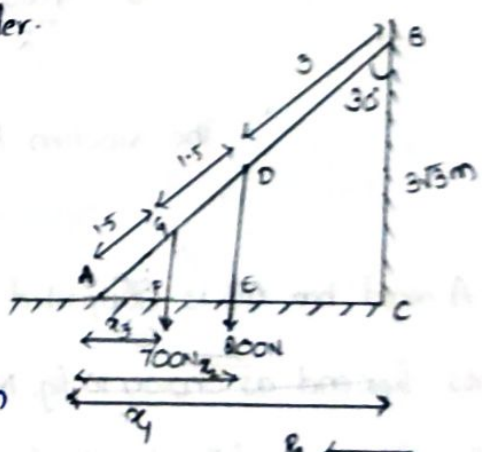
$T(l/2)\cos\alpha = Ql\sin\alpha$

$T = 2Q\tan\alpha$

$\therefore$ Tension in CD $= 2Q\tan\alpha$



14) A uniform ladder AB whose weight is 200N and length 6m rests against a smooth vertical wall making with it an angle of 30°. The other end rests on the ground surface. Find the reactions given to the ladder by the wall and the floor and their inclination to the vertical, when man weighing 700N climbs the ladder by a distance 1.5m along the length of the ladder.

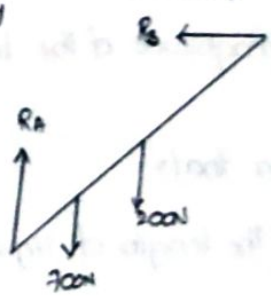


Sol: Given that,

Weight of the ladder $W_L = 200N$

weight of the man $(W_m) = 700N$

Man climbs a distance along the ladder $= 1.5m$



from ΔACB,

$\sin 30° = \frac{x_1}{6}$

$x_1 = 3m$

From ΔADE,

$\sin 30° = \frac{x_2}{3}$

$x_2 = 1.5m$

In Δ AGF,

$\sin 30 = \frac{x_3}{1.5}$

$x_3 = 0.75m$

From free body diagram,

$\Sigma F_y = 0$

$R_A = 200 + 700$

$R_A = 900N$

Consider, $\Sigma M_A = 0$

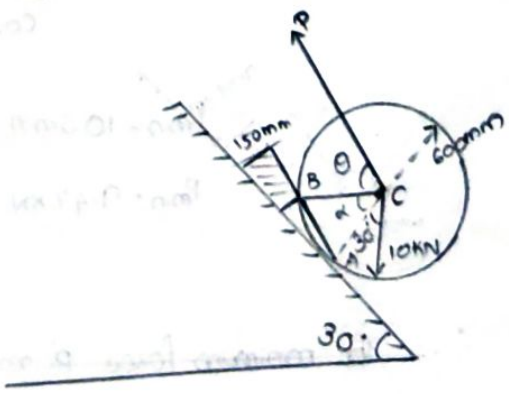
moment at A,

$R_B \times (3\sqrt{3}) - 700(0.75) - 200(1.5) = 0$

$R_B = \frac{700(0.75) + 200(1.5)}{3\sqrt{3})}$

$R_B = 158.771324 N$

15) Determine the magnitude and direction of the smallest force 'P' required to start the wheel figure over the block.



Sol: Given that,

The weight of the wheel = 10KN

radius of wheel = 0.6m

inclination of angle = 30°

Side length of the block = 0.15m

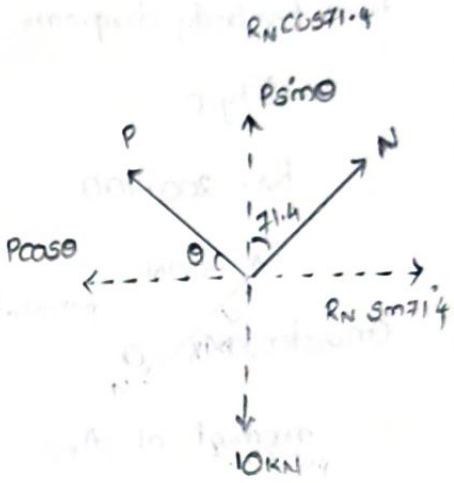
$\cos\alpha = AC/BC = \frac{0.6-0.15}{0.6} = \frac{0.45}{0.6}$

$\cos\alpha = 3/4$

$\alpha = \cos^{-1}(0.75)$

$\alpha = 41.4^\circ$

From free body diagram,



$\Sigma F_x = 0$

$-P\cos\theta + R_N \sin(71.4) = 0$ — ①

$R_N = \frac{P\cos\theta}{\sin 71.4}$

$\Sigma F_y = 0$

$P\sin\theta + R_N \cos 71.4 - 10 = 0$ — ②

$P\sin\theta + \frac{P\cos\theta}{\sin 71.4} \cos 71.4 = 10$

$P\sin\theta \sin(71.4) + P\cos\theta \cos(71.4) = 10 \sin(71.4)$

$P\cos(\theta - 71.4) = 10 \sin(71.4)$

$P = \frac{10 \sin(71.4)}{\cos(\theta - 71.4)}$

$P_{min} = 10 \sin 71.4$

$P_{min} = 9.47$ KN

{For P_{min}
$\cos 0 = 1$
$\theta - 71.4 = 0$
$\theta = 71.4$}

∴ The minimum force 'P' required to start the wheel $P_{min} = 9.47$ KN.

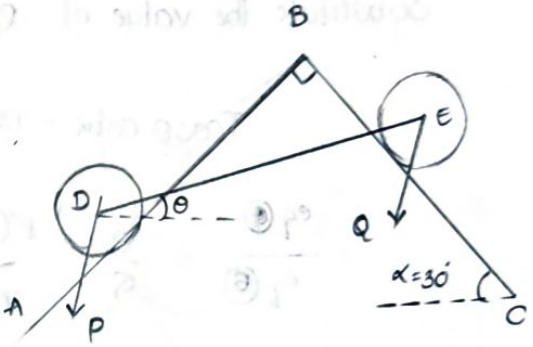
16. Two rollers of weights P and Q are connected by a flexible string DE and rest on two mutally perpendicular planes AB and BC, as shown in fig. Find the tension 'T' in the string and the 'θ' that it makes with the horizontal when the system is in equilibrium. The following numerical data are given. P = 270N, Q = 450N, α = 30°. Assume that the string is inextensible and passess freely through slots in the smooth inclined planes AB and BC.

Sol: Given that,

Weight of D roller (P) = 270N

weight of E roller (Q) = 450N

Angle inclined at C = (α) = 30°

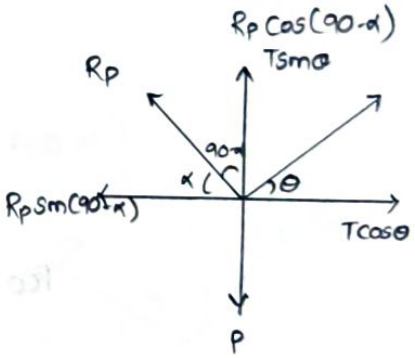


Free Body Diagram at 'P',

$\Sigma F_x = 0$

$T\cos\theta = R_p \sin(90-\alpha)$

$T\cos\theta = R_p\cos\alpha$ — (1)



$\Sigma F_y = 0$

$R_p \cos(90-\alpha) + T\sin\theta - P = 0$

$R_p\sin\alpha + T\sin\theta - P = 0$ — (2)

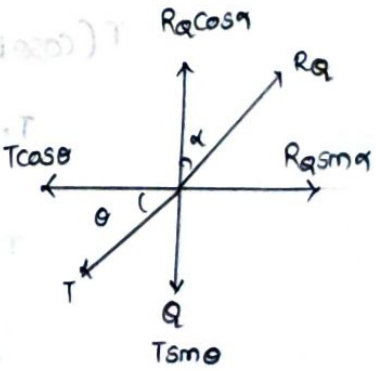
From Free Body Diagram, at Q

$\Sigma F_x = 0$

$R_Q\sin\alpha = T\cos\theta$ — (3)

$\Sigma F_y = 0$

$R_Q\cos\alpha = Q + T\sin\theta$ — (4)



From equation ①

$$R_P = \frac{T\cos\theta}{\cos\alpha}$$

From equation ③

$$R_Q = \frac{T\cos\theta}{\sin\alpha}$$

Substitute the value of R_P in eq ②

$$T\cos\theta\tan\alpha - P + T\sin\theta = 0$$

$$P = T\cos\theta\tan\alpha + T\sin\theta \quad \text{— ⑤}$$

Substitute the value of R_Q in eq ④

$$T\cos\theta\cot\alpha - T\sin\theta = Q \quad \text{— ⑥}$$

$$\frac{\text{eq ⑤}}{\text{eq ⑥}} \cdot \frac{P}{Q} = \frac{\cancel{T}(\sin\theta + \cos\theta\tan\alpha)}{\cancel{T}(\cos\theta\cot\alpha - \sin\theta)}$$

$$P/Q = \frac{\sin\theta + \cos\theta\tan\alpha}{\cos\theta\cot\alpha - \sin\theta}$$

(Take $\cos\theta$ common from numerator and denominator)

$$P/Q = \frac{\tan\theta + \tan\alpha}{\cot\alpha - \tan\theta}$$

$$P\cot\alpha - P\tan\theta = Q\tan\theta + Q\tan\alpha$$

$$\tan\theta = \frac{P\cot\alpha - Q\tan\alpha}{P + Q} = \frac{270\sqrt{3} - 450(1/\sqrt{3})}{270 + 450} \qquad (\alpha = 30^\circ)$$

$$\tan\theta = 0.288$$

$$T(\cos\theta\tan\alpha - \sin\theta) = P$$

$$T = P/\left(\frac{1}{1.04} \times \left(\frac{1}{\sqrt{3}}\right) - \frac{0.288}{1.04}\right)$$

$$T = \frac{270}{0.27822}$$

$$T = 970.455\,N$$

$\therefore$ Tension in the string $= 970.455\,N$

17, A Ringed bar AB with rollers of weights P = 50N and Q = 100 N and at its ends is supported inside a circular ring in a vertical plane as shown in fig. The radius of the ring and length AB are such that the radii AC and BC from a right angle at 'c' that is, $\alpha + \beta = 90°$. Neglecting friction and weight of the bar AB, find the (horizontal) Configuration of equilibrium as defined by the angle $\frac{(\alpha-\beta)}{2}$ that makes with horizontal. Find also the reactions R_A and R_B and the compressive force 'S' in the bar AB.

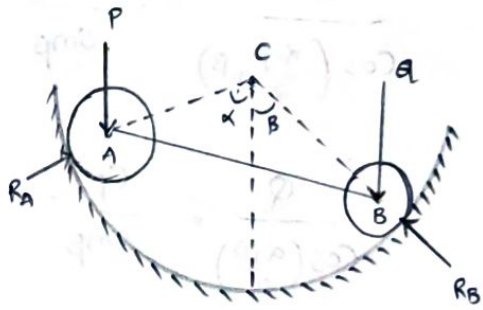
Sol:- Given that,

Weight of roller P = 50N

Weight of roller Q = 100N

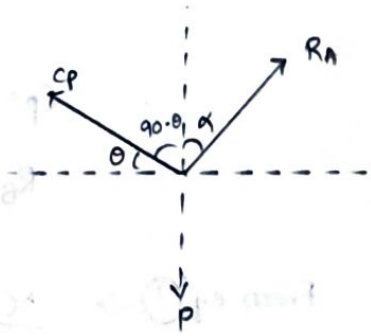
$$\alpha + \beta = 90$$

$$\theta = \left(\frac{\alpha-\beta}{2}\right)$$



By using lami's theorem,

FBD at 'A'



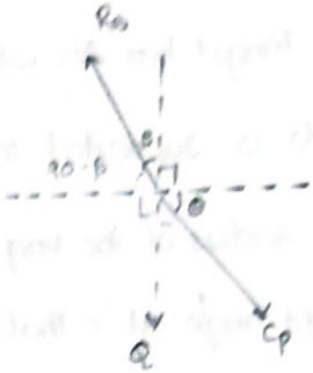
$$\frac{P}{\sin(90-\theta+\alpha)} = \frac{R_A}{\sin(90+\theta)} = \frac{C_P}{\sin(90+90-\alpha)}$$

$$\frac{P}{\cos(\alpha-\theta)} = \frac{R_A}{\cos\theta} = \frac{C_P}{\sin\alpha}$$

$$\frac{P}{\cos\left(\alpha-\left(\frac{\alpha-\beta}{2}\right)\right)} = \frac{R_A}{\cos\left(\frac{\alpha-\beta}{2}\right)} = \frac{C_P}{\sin\alpha}$$

$$\frac{P}{\cos\left(\alpha+\beta/2\right)} = \frac{R_A}{\cos\left(\alpha-\beta/2\right)} = \frac{C_P}{\sin\alpha} \quad \text{--- (2)}$$

FBD at 'B'



By using lamis theorem,

$$\frac{Q}{\sin(90+\beta+\theta)} = \frac{C_P}{\sin(90+90-\beta)} = \frac{R_B}{\sin(90-\theta)}$$

$$\frac{Q}{\cos(\theta+\beta)} = \frac{C_P}{\sin\beta} = \frac{R_B}{\cos\theta}$$

$\alpha+\beta=90^\circ$

$P=50N$

$Q=100N$

$$\frac{Q}{\cos\left(\frac{\alpha-\beta}{2}+\beta\right)} = \frac{C_P}{\sin\beta} = \frac{R_B}{\cos(\alpha-\beta/2)}$$

$$\frac{Q}{\cos\left(\frac{\alpha+\beta}{2}\right)} = \frac{C_P}{\sin\beta} = \frac{R_B}{\cos(\alpha-\beta/2)} \quad -(2)$$

From eq (2) $\Rightarrow$ $\frac{100}{\cos 45^\circ} = \frac{C_P}{\sin\beta} = \frac{R_B}{\cos(\alpha-\beta/2)}$

$$C_P = 100\sqrt{2}\sin\beta$$

$$R_B = 100\sqrt{2}\cos(\alpha-\beta/2) \quad -(3)$$

From eq (1) $\Rightarrow$ $\frac{50}{\cos 45^\circ} = \frac{C_P}{\sin\alpha} = \frac{R_A}{\cos(\alpha-\beta/2)}$

$$C_P = 50\sqrt{2}\sin\alpha$$

$$R_A = 50\sqrt{2}\cos(\alpha-\beta/2) \quad -(4)$$

from (3) and (4)

$$R_B = 2R_A$$

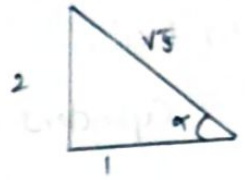
Equating ③ & ④

$\alpha + \beta = 90$

$$100\sqrt{2}\sin\beta = 50\sqrt{2}\sin\alpha \quad -⑤$$

$$2\sin(90-\alpha) = \sin\alpha$$

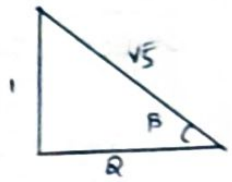
$$2 = \tan\alpha$$



$$\sin\alpha = 2/\sqrt{5} \qquad \cos\alpha = 1/\sqrt{5}$$

Substitute the value of '$\sin\alpha$' in eq ⑤

$$\overset{2}{\cancel{100}}\sqrt{2}\sin\beta = \cancel{50}\sqrt{2}\,(2/\sqrt{5})$$



$$\sin\beta = 1/\sqrt{5} \qquad \cos\beta = 2/\sqrt{5}$$

$$\Rightarrow \beta = \sin^{-1}(1/\sqrt{5})$$

$$\beta = 26.5605$$

$$\Rightarrow \alpha = \sin^{-1}(2/\sqrt{5})$$

$$\alpha = 63.4349$$

$$\Rightarrow \alpha - \beta = 36.8744$$

$$\Rightarrow \theta = \alpha - \beta/2 = 18.4372^\circ$$

$$R_A = 50\sqrt{2}\cos(\alpha - \beta/2) = 50\sqrt{2}\cos(18.4372^\circ)$$

$$R_A = 67.08116$$

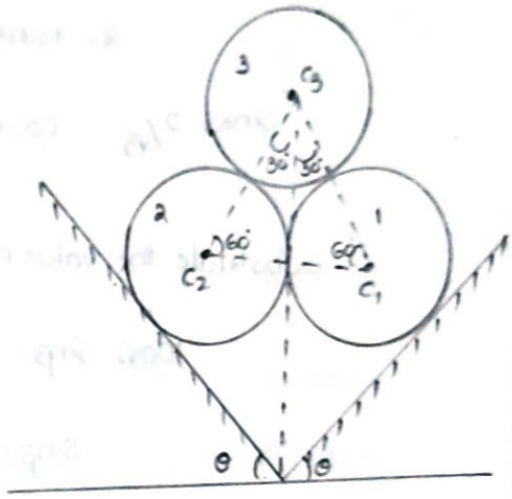
$$R_B = 100\sqrt{2}\cos(\alpha - \beta/2) = 100\sqrt{2}\cos(18.4372)$$

$$R_B = 134.1623$$

18) Three identical cylinders, each weighing w, are stacked as shown in fig, on smooth inclined surfaces, each inclined at angle 'θ' with the horizontal. Determine the smallest angle 'θ' to prevent stack from collapsing.

Sol:- Given that,

Cylinders each weighing 'w'



FBD of C_3

$$\frac{w}{\sin 60^\circ} = \frac{R_{C_2}}{\sin(90+60^\circ)} = \frac{R_{C_1}}{\sin(90+60^\circ)}$$

$$\frac{2w}{\sqrt{3}} = 2R_{C_2} = 2R_{C_1} \quad \text{—(1)}$$

FBD of C_2

$\Sigma F_y = 0 \quad R_A \cos\theta = w + w/\sqrt{3} \cos 30^\circ$

$\Sigma F_x = 0 \quad R_A \sin\theta = w/\sqrt{3} \sin 30^\circ$

$$R_A \cos\theta = w + w/\sqrt{3}\,(\sqrt{3}/2)$$

$$R_A \cos\theta = 3w/2 \quad \text{—(2)}$$

$$R_A \sin\theta = \frac{w}{\sqrt{3}}\left(\frac{1}{2}\right) \quad \text{—(3)}$$

$\frac{(3)}{(2)}$ $\tan\theta = 1/3\sqrt{3} \quad \Rightarrow \theta = 10.8933^\circ$

∴ The smallest angle to prevent stack from collapsing = 10.8933°